

Designed by Remus Teodorescu 2026

LECTURE 12.1

Power Grid Stability Estimation

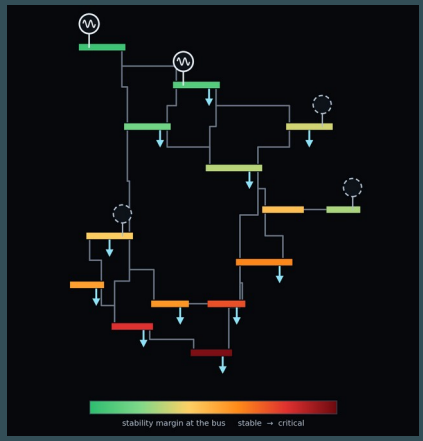
Remus Teodorescu

With support from Research Assistant Noman Khan - nomank@energy.aau.dk

Aalborg University · ret@et.aau.dk

Deep Learning for Engineering · 2026

GRID · STATE · INERTIA



A grid is a graph with the power flow equations at every bus, measured by a few noisy meters. The voltage at every bus is estimated by weighted least squares and by a physics-informed network, with no true state; then a short circuit makes a generator's rotor oscillate, and its inertia is found from the record.

Learning Objectives

SECTIONS, IN ORDER

1. **The grid** — a graph, the power flow equations, what a room sees
2. **State estimation** — least squares, and a network with no true state
3. **The swing** — the equation of motion, and the inertia from a record

UDL carries the graph network, Liu the inverse fit. The example is Ex_12.1's six-bus grid, representative of eastern Denmark and not a model of it: the line impedances are textbook values, because no operator publishes measured ones, and results that depend on them must say so.

READ, AND WHAT EACH ANSWERS

UDL ch. 13	how is a graph represented?
Liu ch. 4	what can readings determine?
SciPy	how is an ODE shot and fitted?

Recap of L11.2 Obstacle Avoidance

1 – 2 · IDENTIFYING AN OBSTACLE

1 **How can a network tell from one frame whether the path is blocked, where a dark-pixel rule fails?**

It has two outputs, free and blocked, turned into probabilities by the softmax and trained with cross-entropy. The rule stops for shadows and misses grey boxes. On the rendered frames the rule was right 85 % of the time, the network 97 %. *Go deeper: the frames each one gets wrong (Ex_11 section 5); UDL ch. 5.*

2 **Why is the threshold on the probability of a blocked path not set at one half?**

A missed obstacle costs a collision, a false stop a second. Lowering the threshold trades misses for false stops; choose it where the expected cost is smallest. At one half the network missed 15 % of the obstacles, at 0.05 it missed 4 % and stopped for 5 % of the free frames. *Go deeper: the cost against the threshold (Ex_11 section 5); measure both rates on the car.*

3 – 4 · DECIDING THE ACTION

3 **How does the avoidance strategy decide what the network must tell, and why does stopping not reach B?**

Stopping needs only free or blocked, but an obstacle that stays keeps the car standing. Steering around needs which side is free; a potential field needs where each obstacle is. On the course the field reached B in 6.3 s, steering around in 8.3 s. *Go deeper: the three strategies (Ex_11 section 6); UDL ch. 19.*

4 **Why does the rate of the control loop limit the safe speed of a car that avoids obstacles?**

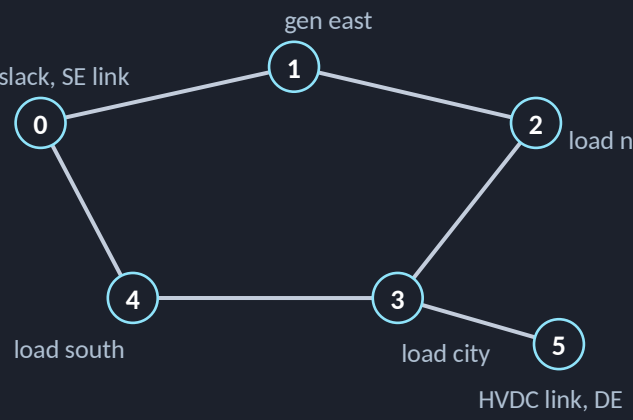
From the frame that shows the obstacle, the car drives one blind distance, then brakes as far as its tyres allow, and both must fit within what the camera sees. With 0.7 m that is 1.9 m/s at 5 decisions a second and 2.7 m/s at 30. *Go deeper: the safe speed (Ex_11 section 7); what a faster network shortens, and what not.*

A Grid Is a Graph

Transmission grid — buses joined by lines: a graph, with a node's state being its voltage magnitude and angle, and every edge carrying an admittance. The state of the whole grid is one number pair per bus, and everything a control room wants to know is a function of it.

EX_12.1'S SIX BUSES, AS THEY ARE WIRED

SIX BUSES, SIX LINES, ONE RING



a bus is a node, numbered as in the notebook

each line is an edge, weighted by its admittance $G + jB$

a ring of five buses, and one radial line to the HVDC bus

THE CASE, AND ITS LIMITS

the grid	6 buses, 6 lines, 100 MVA base
the state	6 magnitudes, 5 angles
stands for	eastern Denmark, loosely
impedances	textbook values, flagged
so	say when a result leans on them

No operator publishes measured impedances; representative is the honest word.

The Power Flow Equations

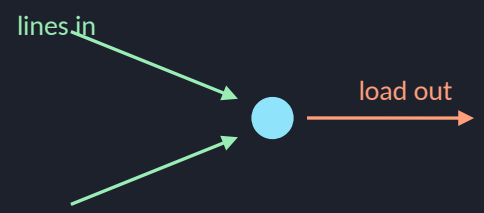
Power flow equations — two algebraic balances at every bus: the active and the reactive power the lines bring must equal what the bus injects or takes. They are constraints the voltages satisfy at every instant, not a simulation in time - the elliptic habit of L8.1, on a graph.

TWO BALANCES AT EVERY BUS

$$P_i = V_i \sum_k V_k (G_{ik}\cos\theta_{ik} + B_{ik}\sin\theta_{ik}), \qquad Q_i = V_i \sum_k V_k (G_{ik}\sin\theta_{ik} - B_{ik}\cos\theta_{ik})$$

V_i, θ_i the voltage magnitude and angle at bus i : the state, G, B the admittance matrix, $\theta_{ik} = \theta_i - \theta_k$

ONE BUS, ONE BALANCE



what the lines bring

equals what the bus takes

two balances: P and Q

THREE WAYS THE EXERCISE USES THEM

1 · solver	Newton, all injections given: the truth
2 · check	mismatch ΔP, ΔQ of any estimate, per bus
3 · loss	minimise the mismatch, in torch

Why a loss: in operation nobody knows the true state, so there is no label to compare an estimate with. The physics can still judge it, since the true state leaves zero mismatch wherever the injection is known: at a meter, a forecast or a bus with nothing connected. Written in torch, the mismatch passes its gradient back through the voltages to the weights.

What the Control Room Sees

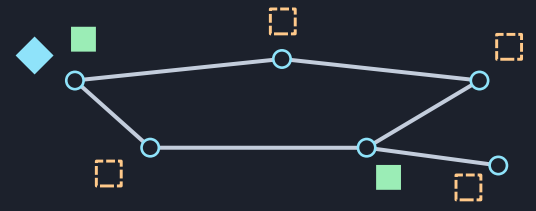
Measurement model — every reading is a function of the state plus noise: a PMU reads a bus's magnitude and angle, a meter its P and Q, a forecast stands in where no meter exists. The thin set has fewer readings than unknowns, and where they sit on the graph decides everything.

ONE LINE PER METER

$$z_j = h_j(V, \theta) + \varepsilon_j$$

z_j one reading, h_j what that meter measures as a function of the state, ε_j its noise

THE FORECAST SET ON THE GRID



PMU: $|V|$ and θ at its bus
meter: P, Q injected there
forecast stands in, loosely
shown: the forecast set, the set the network uses

TALK · 2 MIN How would you explain, in your own words, why a grid's state can be known better than its meters? Then, from the table on the right, why does the thin set fail where the forecast set does not?

THE THREE SETS OF EX_12.1

set	readings	covers
full	14 = 4 + 10	every unknown
thin	6	5 of 11 unknowns
forecast	6 + priors	every unknown, loosely

Full: PMUs at buses 0 and 3 read V, θ(4), meters at buses 0 – 4 read P, Q(10): 14 readings for 11 unknowns. Bus 5 has no meter, yet the balance at bus 3 sums over its neighbours, so P_3, Q_3 fix bus 5 too. Coverage is where the meters sit on the graph, not how good they are.

Weighted Least Squares

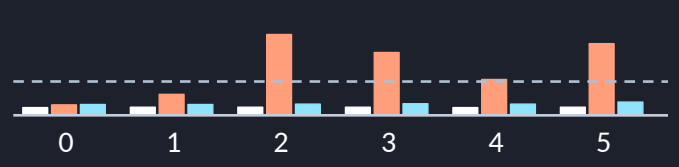
State estimation — finding the voltage magnitude and angle at every bus from fewer, noisy readings. The estimator every utility runs is weighted least squares: each meter's misfit, scaled by its noise, minimised by Gauss–Newton iterations. It is the baseline any network must face.

THE ESTIMATOR IN SERVICE

$$\hat{x} = \arg\min_{x} \sum_{j} \frac{(z_j - h_j(x))^2}{\sigma_j^2} \qquad \sigma_j^2 = \mathrm{E}\left[\varepsilon_j^2\right]$$

z_j the reading of meter j, $h_j(x)$ what meter j would read if the state were x, $\hat{x}$ the estimated state
σ_j the standard deviation of the noise ε_j of meter j, $\mathrm{E}[\,\cdot\,]$ the expected value : the mean over many readings

ERROR IN THE VOLTAGE MAGNITUDE AT EACH BUS



full set, 14 readings
thin set, 6: worst at buses 2, 5, 3
forecast set
dashed: the target, 0.01 p.u.

Why it works: each reading is an equation in V, θ through h_j. With independent normal noise the weighted sum is minus twice the log-likelihood, so $\hat{x}$ is the most probable state.

WORST BUS ERROR, MEAN OF 300 POINTS

set	in $\lvert V \rvert$	angle	worst bus
full, 14	0.0030 p.u.	0.07°	5, the HVDC
thin, 6	0.0248 p.u.	1.82°	2, load north
forecast	0.0050 p.u.	0.15°	5, the HVDC

Each value is the error at the worst bus of an operating point, averaged over 300 points. The exercise sets its target at 0.01 p.u., every bus within 1 % of its nominal voltage. One estimate takes about 2 ms on a laptop processor: eleven unknowns and a few iterations. The thin set fails because its six readings do not determine six of the eleven unknowns.

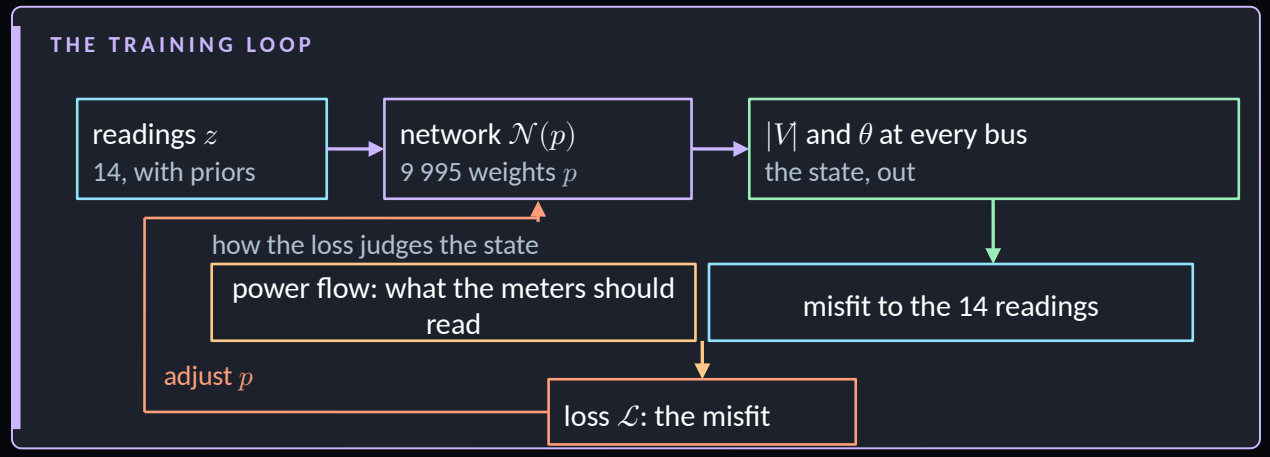
A Network Constrained by Physics

Physics-constrained estimation — a second estimator: a network maps the readings to the state at every bus. Its loss is the weighted misfit of least squares, averaged over many operating points, with the power flow equations inside h_j. No true state appears anywhere in the loss.

THE TRAINING LOSS: WEIGHTED MISFIT, NO TRUE STATE

$$\mathcal{L} = \frac{1}{N}\sum_{n}\sum_{j}\frac{\left(z_j^{(n)} - h_j(\hat{x}^{(n)})\right)^2}{\sigma_j^2}, \qquad \hat{x}^{(n)} = \mathcal{N}(z^{(n)}; p)$$

$\mathcal{N}$ the network, p its weights, n one of N operating points; h_j of a P or Q reading is the power flow equations a bus with no meter enters through a forecast of its injection, as a reading with a large σ_j; no true state in the loss



AGAINST THE BASELINE, HONESTLY

accuracy	0.0043 p.u.: it does not win
one estimate	0.26 ms; 2 µs in a batch
training	13 s, once
a line out, untold	both fail alike
but the check	its misfit J: 0.28 to 2.35

The gain is the time of one estimate, and a self-check that says when not to trust it.

The Swing Equation

Swing equation — the grid set in motion: a machine's rotor accelerates with the gap between its mechanical and its electrical power, held back by inertia and damping, and the electrical power ties the machines together. One ordinary differential equation per machine.

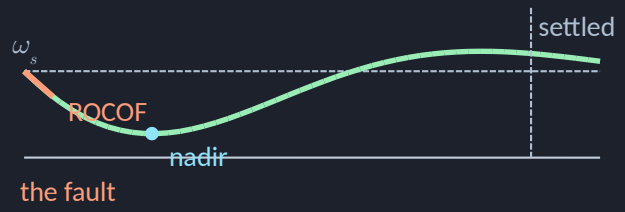
THE EQUATION OF MOTION

$$\frac{d\delta}{dt} = \omega - \omega_s, \qquad \frac{2H}{\omega_s}\frac{d\omega}{dt} = P_m - P_e(\delta) - D\frac{\omega - \omega_s}{\omega_s}$$

δ the rotor angle, ω its speed, H the inertia in seconds, D the damping, P_m, P_e mechanical and electrical power

Right after a fault the speed has not yet moved, so only H acts: the frequency first falls at ROCOF $= f_s\ \Delta P/2H$. Once the speed has left ω_s, D brakes the swing; for one machine against a stiff grid it dies away with time constant $4H/D$.

AFTER A FAULT



ROCOF: how fast it first falls
nadir: the deepest deviation
settling time: until it dies away
more renewables: H falls, the dip faster and deeper

INERTIA H AGAINST DAMPING D

	inertia H	damping D
what it is	stored spin energy	a brake on ω - ω_s
unit	seconds	p.u. per p.u. speed
comes from	rotating masses	dampers and loads
resists	any change of speed	a speed off ω_s
sets in the trace	ROCOF and the nadir	how fast the swing dies
Ex_12.1's plant	4.0 s	1.2
with converters	falls	set by their control

H is the kinetic energy of the rotor at rated speed divided by its rating: the seconds the machine could deliver rated power from its spin alone. H and D act at different moments of the swing, so one recorded disturbance fixes both. Converters have no rotating mass: H falls as they replace machines.

The Inertia From a Record

Inertia identification — finding the inertia H and the damping D of a machine from a recorded disturbance. Both are trained together with the network's weights until the swing equation and the record agree, as the diffusivity was in L8.2. Only the record after the fault is cleared is used.

THE LOSS: SWING EQUATION PLUS RECORD

$$\mathcal{L} = \frac{1}{N}\sum_k \left(\frac{2H}{\omega_s}\dot{\omega}_k - P_m + P_{e,k} + D\,\Delta\omega_k\right)^2 + w_D \frac{1}{M}\sum_m (\omega_m - \omega_m^{rec})^2$$

k one of N instants after the fault, $\dot{\omega}_k$ the network's acceleration by autograd, $\Delta\omega_k = \omega_k - \omega_s$
ω_m^{rec} the recorded speed at M instants, w_D its weight; H, D trained with the weights

In words: the first sum asks the network's angle and speed to obey the swing equation at many instants after the fault; the second asks its speed to match the recorded frequency. H and D are the only physical numbers allowed to change, so the values that satisfy both are the machine's. H multiplies the acceleration: had it divided it, a very large H would shrink the first sum and pass for a good fit.

TALK · 2 MIN How would you decide, in your own words, which part of a record identifies a parameter? Then, from the table on the right, why does the record before the fault fail from both first guesses?

INERTIA FOUND, TRUE VALUE 4.00 s

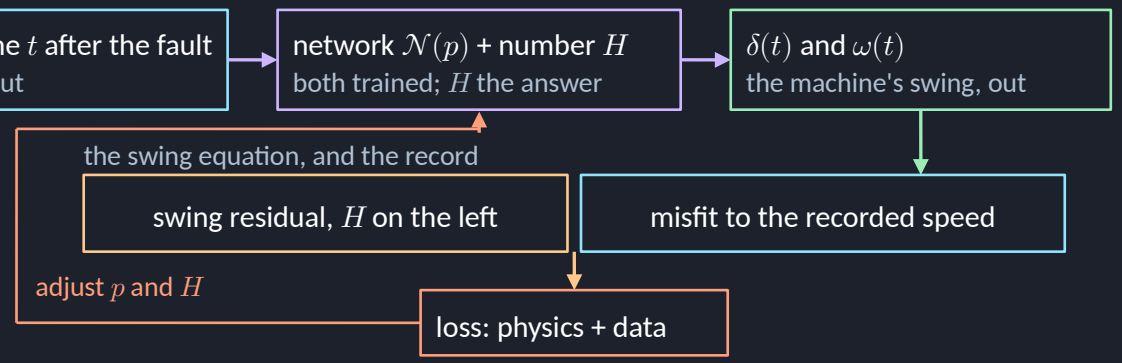
method, part of the record	first guess 8 s	first guess 2 s
shooting, after the fault	7.41 s	1.83 s
PINN, after the fault	3.89 s	3.90 s
shooting, before the fault	50 s, its limit	1.94 s
PINN, before the fault	8.35 s	2.10 s

Shooting is the classical fit: simulate the swing equation with a guessed H, compare with the record, correct the guess, repeat. Each method starts from a first guess, 8 s or 2 s; a result that depends on it has stopped in a local minimum. After the fault the PINN finds 3.9 s from both. Before the fault nothing disturbs the machine, so the record carries no information on H and no method can find it.

Solving the Inverse Problem

Inverse training loop — the previous slide's fit as a training loop: the time since the fault in, the angle and the speed out, checked by the swing equation and the record; the weights, H and D adjust together.

THE INVERSE LOOP



THE KEY CALLS IN PYTORCH

```
log_H = nn.Parameter(torch.tensor(8.0).log())  # H as log H: positive
opt = torch.optim.Adam([*net.parameters(), log_H, log_D])  # all trained
dw = torch.autograd.grad(w.sum(), t, create_graph=True)[0]  # dω/dt by autograd
res = 2*log_H.exp()/w_s*dw - P_m + P_e + log_D.exp()*(w - w_s)/w_s  # swing eq.
loss = res.pow(2).mean() + w_D*(w_m - w_rec).pow(2).mean()  # equation + record
opt.zero_grad(); loss.backward(); opt.step()  # one step for weights, H and D
```

WHAT GOES IN AND WHAT COMES OUT

data used	the recorded speed in the 0.8 s after the fault is cleared
trained	the 9 995 weights of the network, plus H and D
physics	the swing equation in the loss, H multiplying the acceleration
result	H = 3.89 s against the true 4.00 s: 3 % low
kept	only H and D; the network is then discarded

The network only gives the swing a smooth, differentiable form in time. What the training delivers is the machine's inertia.

Advanced Topics

RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

topic	what it adds	read	where it makes a difference
bad-data detection	a wrong meter, found by its residual	Liu ch. 4	every estimator in service
placing one more meter	where a reading buys observability	pandapower docs	the thin set's missing half
the critical clearing time	how long a fault may stay	PowerSystems.jl	protection settings
rate of change of frequency	the first seconds of a real record	ENTSO-E data	low-inertia grids
the graph network's layers	messages passed along the lines	UDL ch. 13	grids too large for one MLP; L12.2

WHY THEY ARE LEFT OUT

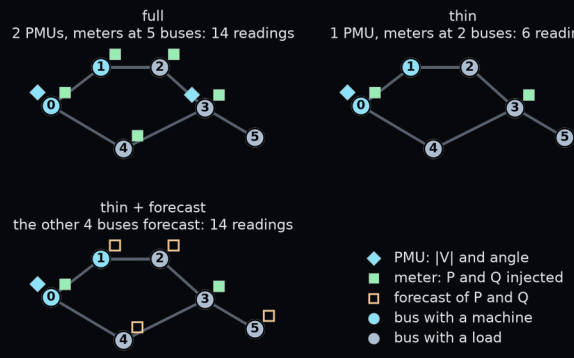
Ex_12.1 estimates one six-bus grid from fourteen readings and fits one machine's inertia, scored against Newton's power flow, so none of them is needed to do it. The last row is where the next lecture begins.

EXERCISE

Exercise 12.1

ONE NOTEBOOK, SECTION BY SECTION

- **1 – 3** — the grid, the three sets of readings, the power flow
- **4** — Newton–Raphson reference; least squares on each set
- **5** — the physics-informed network; then a line out, untold
- **6** — the inertia from the recorded disturbance, two ways
- **7 – 9** — the report; mini projects MP12.1A and MP12.1B



THE QUESTION EACH PART ANSWERS

1 – 3	what is the state of a grid?
4	how far do the meters carry?
5	what replaces the true state?
6	what does a record identify?

REPORT QUESTION, TO CONCLUDE

The network matched least squares from the same readings and answered three hundred estimates in a blink; its residual rose eightfold when a line went out untold. In a control room, what would you use each tool for?

SUMMARY

Key Takeaways

STATE ESTIMATION	
what it is	the state from few, noisy meters
function	every bus's V and θ
train, tune	no true state: physics in the loss
vs least squares	equal accuracy, a fraction of the time
used for	a room's picture, and its self-check

SWING DYNAMICS	
what it is	the grid's equation of motion
function	angles and speeds after a fault
train, tune	H, D beside the weights
vs shooting	no local minima on this record
used for	today's inertia, from records

Next — L12.2: whether the grid survives the next fault, predicted before it happens.

Questions

1 – 2 · A NETWORK FOR THE STATE

1 **How can a network estimate the state of the grid from the readings when no true state exists for training?**

It takes the fourteen readings and returns the voltage and angle at every bus. The power flow equations turn that state into what each meter should read, and the loss is the weighted misfit to the actual readings. No true state appears in the training. *Go deeper: the loss (Ex_12.1 section 5); UDL ch. 13.*

2 **Weighted least squares is already a statistical estimator: what does a network add to it?**

Both minimise the same weighted misfit, so from the same readings they are equally accurate. Least squares solves it anew for every set of readings by Newton iterations; the network learns it once. Its gain is time, a quarter of a millisecond, and its misfit flags a line out. *Go deeper: the line-out test (Ex_12.1 sections 4 – 5); Liu ch. 4.*

3 – 4 · THE INERTIA FROM A RECORD

3 **How can the inertia of a machine be found with a network from a recorded disturbance?**

The network takes the time after the fault and returns the angle and the speed; the inertia is one more trainable number. The loss is the residual of the swing equation plus the misfit to the record. The inertia must multiply the acceleration; dividing by it lets a huge value hide. *Go deeper: the inverse problem (Ex_12.1 section 6); Liu ch. 4.*

4 **Which part of a record identifies the inertia, and how does the network fit compare with the classical one?**

Only the swing after the fault; before it nothing moves, and every fit stays near its first guess. After the fault the network fit found 3.89 s from both first guesses against the true 4.00 s, while the classical fit stayed near its guess. *Go deeper: fit both windows (Ex_12.1 section 6); why two first guesses must agree.*

NEXT

L12.2 — Power Grid Stability Prediction

From what the grid is to what it will do: whether it survives the next fault, predicted by a graph network in the time a protection relay has.

BEFORE THE NEXT SESSION

read	UDL ch. 13
run	Ex_12.1, sections 1 to 6
bring	your inertia, from both starts
think about	what the residual map told you

The estimation half of the block is done: the state, and the machine behind it.